K-Youth Training Program

Data Analytics for business growth and sustainability in the workforce

Introduction

Who am I

- Kassim Abdullah @ Vignes

Profile

- Industry expert who has worked in Healthcare, Government, Finance, Telecommunications, Plantation and Manufacturing for over 30 years.
- Designing, developing and deploying systems across industries and countries.
- Specialized in solutions that are data driven and implementation of transformational solution for clients

Ice Breaker

- Introduce yourself
  - Your name (how do people refer to you)
  - Which School of Study / Major
  - What year student
  - Something unique about yourself that not many people know!

- Knowledge check
  - Next slide (Slido)

slido

Please download and install the Slido app on all computers you use

What is Data Analytics (your thoughts)?

ⓘ Start presenting to display the poll results on this slide.

Course Outline

Module 0: Configuration and setup

- 1. PowerBi installation
- 2. Python installation
- 3. Visual Studio Code installation
- 4. Neo4j – graph database

Module 1: Introduction to Data Analytics

- 1. What is data and its importance
- 2. Types of Data Analysis
- 3. Roles in Data
- 4. Real world examples of data and its impact
- 5. Fact Check

Module 2: Data Analysis Fundamentals

- 1. Databases – types and purpose
- 2. Tools used to ingest data
- 3. Storing and retrieving data
- 4. Data Quality and Cleaning Techniques
- 5. Data Transformation
- 6. Data Integration and ETL Processes
- 7. Python Basics for Non-IT Students
- a. Introduction to Python and Its Uses in Data Analytics
- b. Setting Up Python Environment
- c. Basic Python Syntax and Operations
- d. Data Structures: Lists, Tuples, Dictionaries
- e. Importing and Using Libraries (NumPy, Pandas)
- f. Basic Data Manipulation with Pandas

Module 3: Exploratory Data Analysis

- 1. Descriptive Statistics
- 2. EDA Techniques
- 3. Identifying Trends and Patterns

Course Outline

Module 4: Advanced Data Analysis Techniques

- 1. Big Data Analytics
- 2. Text and Sentiment Analysis
- 3. Advanced Machine Learning Techniques

Module 5: Data Modeling and Relationships

- 1. Data modeling introduction
- 2. Why is data modeling important
- 3. Introduction to Microsoft Power BI
- 4. Power BI Interface and Features
- 5. Connecting to Data Sources
- 6. Creating Basic Visualizations in Power BI
- 7. Data Modeling with Power BI
- 8. Graph Databases Introduction
- 9. Importance and growth of Graph Databases
- 10. Cypher basics for querying graph databases
- 11. Modeling a graph database
- 12. Visualization and analytics using a graph

Module 6: Advanced Visualization and Dashboarding

- 1. Advanced Analytics in Power BI
- 2. Advanced Visualization in Power BI
- 3. Python Tools for visualization
- 4. Dashboard concepts and design
- 5. Developing Dashboards in Power BI

Module 7: Real-world Applications and Projects

- 1. Data Privacy and Security
- 2. Ethical Considerations in Data Analytics
- 3. Regulatory Compliance
- 4. Recap of Key Learnings
- 5. Future Trends in Data Analytics

Key Objectives

a) Understand what is data analytics

b) Explore the tools used in the processing of data

c) Able to create / load / analyze data from multiple data sources

d) Generate analysis of the data using tools

e) Understand the visualization and story telling

f) Display skills to provide effective visualizations to business for decision making

g) Get industry certifications that hold value in the workplace

Tools for Data Analysis (Course Prerequisites)

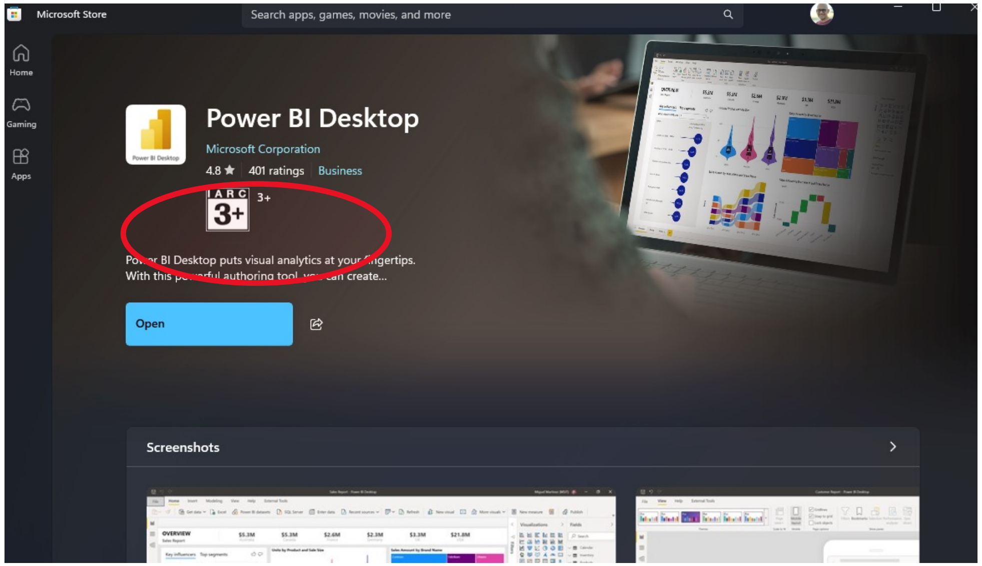
Tools Required

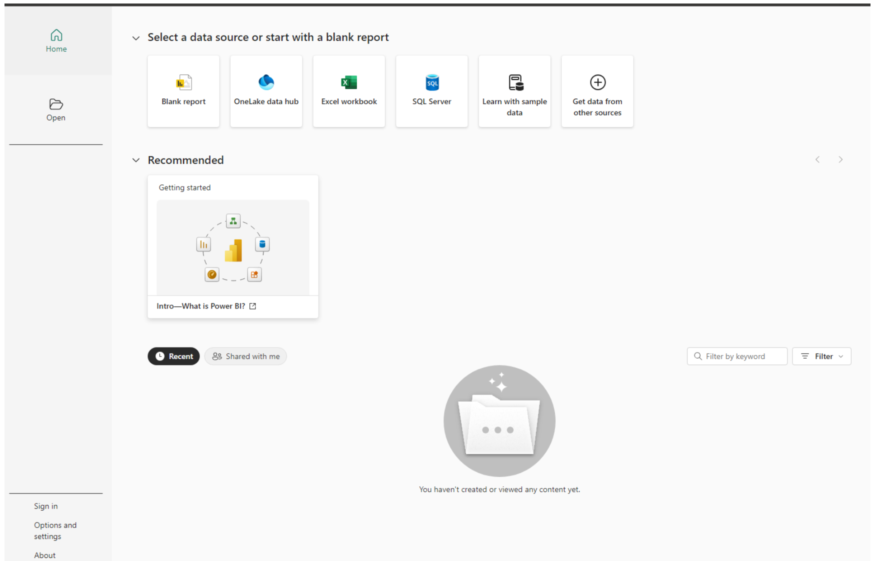
- Python 3.13 - https://www.python.org/downloads/
- Visual Studio Code - https://code.visualstudio.com/
- Microsoft PowerBI Desktop - https://www.microsoft.com/en-us/power-platform/products/power-bi/desktop
- Neo4j Database - https://neo4j.com/download/

Sample Data

- Microsoft Sample Data - https://learn.microsoft.com/en-us/power-bi/create-reports/sample-datasets

Power BI Desktop Download





What is Data?

Data is everywhere !!!

Business and Operations

- CRM – Customer interactions, sales data
- ERP - Financial data, inventory
- POS – Transaction data, e-Commerce transaction
- Logistics – GPS data, tracability, routing information

Web / Social Media

- User Behavior on sites,
- Page visit information
- Time spent on a page
- SEO data
- Google Analytics
- Cookie data
- Search query information
- Location information

IoT / Sensor

- Smart Home
- Wearables
- Industrial Sensors
- GPS tracking
- Traffic data
- Google Maps

Cloud Data

- API access statistics
- Hosted data
- Uploaded data
- Downloaded data
- Bandwidth data

Communication / Collaboration

- Call Logs
- CDR data
- Email data – messages, attachments
- Customer support
- Zoom / Teams etc.

Multimedia Data

- Audio files
- Video Files
- Image files
- Lidar Data
- Hyperspectral Data

Economic and Market

- Stock Market transaction
- Volume of lots
- Cryptocurrency
- Digital Signing
- Economic data – GDP, inflations etc.

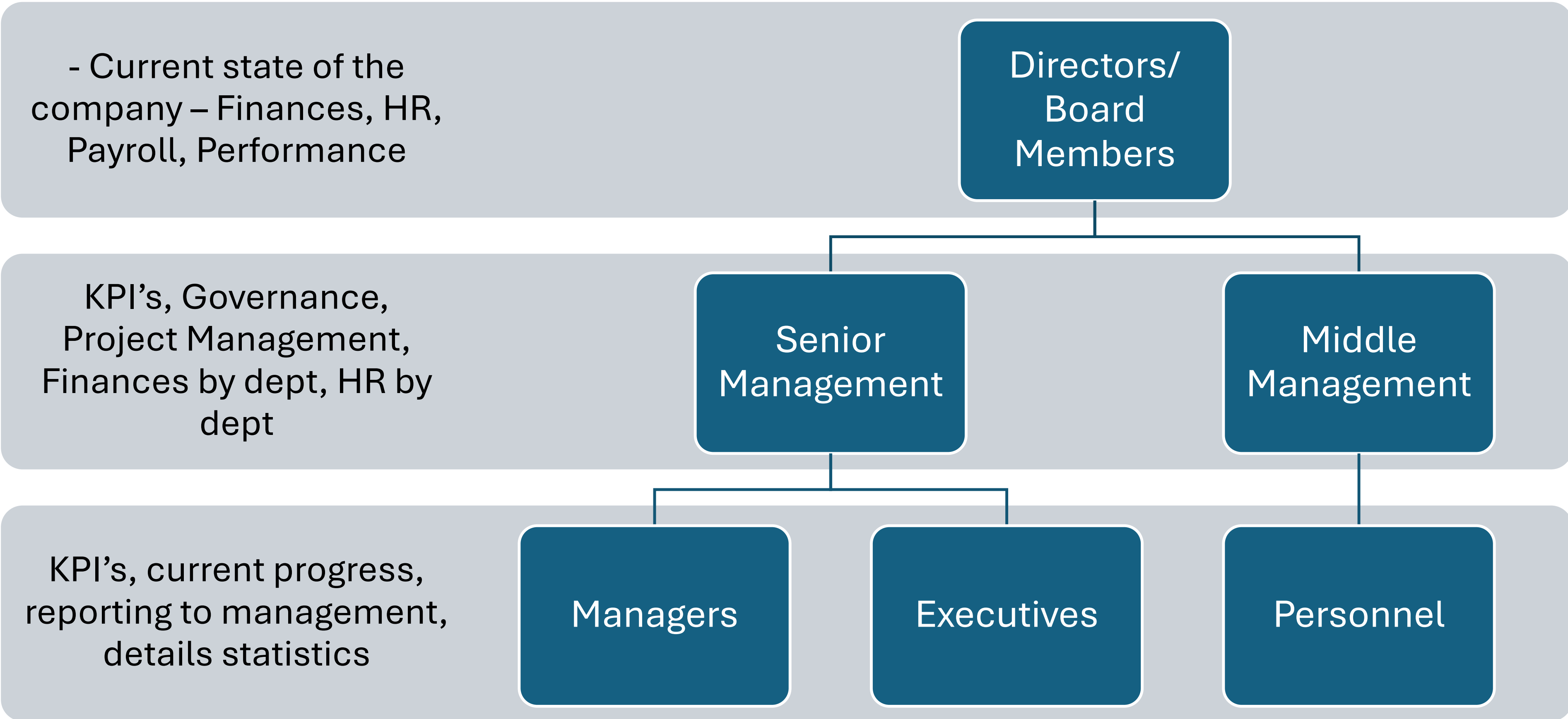
Health

- Wearables
- Lab equipment
- MRI
- CT Scan
- Heart Monitor

Why do we need data?

- Informed Decision Making
- Improve Efficiency and Productivity
- Enhanced Customer Understanding
- Risk Management
- Competitive Advantage

Who uses the data?



Roles in Data

- Data Analyst
- Business Analyst
- Data Scientist
- Data Engineer
- Machine Learning Engineer
- Data Architect
- Data Visualization Specialist
- Data Governance Specialist
- Chief Data Officer

Real life use cases

Netflix operates in a highly competitive streaming market. One of its core objectives is to keep users engaged and reduce churn (customers leaving the platform). The key to achieving this is understanding user preferences and delivering personalized recommendations.

Data:

1. **Data Collection**:
    - Tracks user behaviors, such as:
    - Movies/shows watched.
    - Viewing times and durations.
    - Ratings provided by users.
    - Interactions like searches and clicks.
    - Collects metadata about its content (genres, actors, directors).
2. **Analysis and Insights**:
    - Analyzes viewing patterns to determine:
    - Popular genres.
    - Preferred content formats (e.g., series vs. movies).
    - Trends in different geographic regions.
    - Identifies when users are most likely to stop watching (churn risk).
3. **Decision-Making**:
    - Personalizes user recommendations through algorithms like collaborative filtering and deep learning.
    - Creates content tailored to user preferences, leveraging insights (e.g., producing original series like *Stranger Things* based on demand for sci-fi).
    - Designs marketing strategies targeted to specific audience segments.

Outcomes:

- **Improved Customer Experience**: Data allows Netflix to deliver content users love, enhancing satisfaction and loyalty.
- **Cost Efficiency**: Data helps Netflix decide what content to invest in, avoiding costly flops.
- **Competitive Advantage**: Insights from data help Netflix stay ahead in a crowded market.

Real life use cases

Amazon is one of the largest e-commerce platforms in the world. To maintain its competitive edge, the company must ensure that products are always available for customers, deliveries are fast, and operational costs remain low.

Data:

1. **Data Collection**:
   - Gathers real-time data on:
   - Customer orders and preferences.
   - Product availability and stock levels.
   - Supplier performance and lead times.
   - Shipping and delivery logistics.
   - Uses historical sales data to predict demand for various products.
2. **Analysis and Insights**:
   - Analyzes patterns to forecast demand for specific items across different regions.
   - Optimizes warehouse locations and inventory distribution based on regional purchasing trends.
   - Tracks delivery routes and driver performance to identify inefficiencies.
3. **Decision-Making**:
   - Implements dynamic inventory management systems to stock high-demand items closer to customers.
   - Uses predictive analytics to reduce overstocking or understocking.
   - Optimizes delivery routes with algorithms to minimize delivery times and costs (e.g., Amazon Prime's one- and two-day delivery promises).
   - Personalizes product recommendations to drive more sales.

Outcome:

- Efficiency Gains: By leveraging data, Amazon minimizes waste and maximizes productivity.
- Scalability: Data allows Amazon to scale its operations globally while maintaining consistent service quality.
- Customer Loyalty: Fast delivery and personalized shopping experiences strengthen customer trust and loyalty.